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Meta-Labeling for Multi-Asset Model Position Sizing

Project Walkthrough — Review and Brainstorming

0.83	1.35	10.84%	6.6%
IR	Sharpe	Ann. Return	Ann. Vol

Two-Layer Architecture — Signal Generation & Confidence Sizing

Data Pipeline — Raw Bloomberg to Clean Returns

Data Sources

4 Bloomberg Excel files — SPX, BCOM, 10Y Treasury, IG Corporates. Monthly data, Feb 1996 – Dec 2025 (360 months).

Standardisation

Price series (SPX, BCOM, IG) → simple monthly returns. Treasury yield → total return proxy via duration-adjusted price return + carry.

Output

Single clean returns DataFrame: 360 rows × 4 columns.

Mentor Q: No transaction costs or liquidity constraints assumed. Could these omissions flip the strategy's edge entirely, not just shave returns?

Pipeline Output

```
def compute_bond_total_return(yield_pct, dur=8.5):
 y = yield_pct / 100.0
 dy = y.diff() # Δyield month-over-month
 px = -dur * dy # duration × Δyield
 carry = y.shift(1)/12 # coupon accrual
 return px + carry

# Output: 4 aligned monthly return series
# Feb 1996 – Dec 2025 (360 months)
# No manual data overrides.
```

Indicator Selection — Why These Five?

5 macro indicators — a **minimal, non-redundant set** spanning major macro regimes.

Indicator	Macro Dimension	Economic Logic
SPX Trend	Equity momentum	Price > 6mo avg → expansion signal
Credit vs Rates	Credit risk appetite	Corp bonds outperform UST → risk-on
Risk Breadth	Broad risk participation	Multi-asset > UST → broad rally
BCOM Accel	Commodity inflection	3mo avg − 12mo avg → turning point
Yield Mom	Rate cycle direction	−ΔYield(3mo) → falling rates signal

Mentor Q: BCOM Trend was dropped as redundant with BCOM Accel. Was this correct, or are we leaving a meaningful regime signal on the table?

M1 Indicator Construction

Expanding Z-Score Design

Each indicator → expanding z-score using only past data (min. 12 observations), avoiding look-ahead bias.

Why Expanding, Not Rolling?

No natural stationarity window — a 1998 yield extreme may be normal in 2020. Rolling z-scores assume only the last N months matter, without empirical basis.

> **Mentor Q:** 6mo trend and 3mo relative windows chosen for economic plausibility. Do different window pairs meaningfully change signal quality, or are results robust?

```
def expanding_zscore(series, min_periods=12):
    history = series.shift(1)   # past data only
    mu  = history.expanding(
            min_periods).mean()
    sig = history.expanding(
            min_periods).std()
    return (series - mu) / sig

# Indicator construction:
# Trend:       price / rolling_mean(6mo) - 1
# Credit/Rates: rolling_mean(3mo) corp−ust
# Risk Breadth: rolling_mean(3mo) risky−ust
# BCOM Accel:   rolling_mean(3mo)−mean(12mo)
# Yield Mom:    −Δyield(3mo)
```

Window Rationale

6-month trend: filters monthly noise while remaining responsive within a business cycle phase. **3-month relative:** one quarter smooths idiosyncratic bond market noise without losing regime sensitivity.

Dynamic Information Coefiicient Weighting

The Core Idea

36-month trailing Spearman IC between each indicator's z-score and forward returns. Positive IC → weighted; zero/negative IC → zeroed out. Weights renormalised monthly.

Why Spearman? Why 36 Months?

Indicator predictive power is non-stationary. 36 months captures regime shifts without overfitting short-term noise. Spearman is robust to outliers and non-linear relationships.

Weighting Scheme	Sharpe	IR	Ann. Return
Dynamic IC (ours)	**1.35**	**0.83**	**~11%**
Equal Weight	1.16	0.41	~9.2%
Ranking-Based	1.20	0.45	~9.6%
Walk-Forward OOS	1.10	0.38	~8.8%

```
ic_window = ic_data.iloc[i - 36 : i + 1].dropna()
if len(ic_window) >= 12:
    corrs = ic_window.corr(method="spearman")["target_ret"].drop("target_ret")
    ic_mask = (corrs > 0.0).astype(float)
base_w = inv_corr / inv_corr.sum()   # inverse-correlation weights
final_w = base_w * ic_mask           # ← zero out negative-IC indicators
if final_w.sum() > 0:
    final_w = final_w / final_w.sum()  # re-normalise
```

Mentor Q: Binary on/off by IC sign — should we weight proportionally to IC magnitude instead?

M1 Signal-to-Allocation Mapping

Signal Mapping Logic

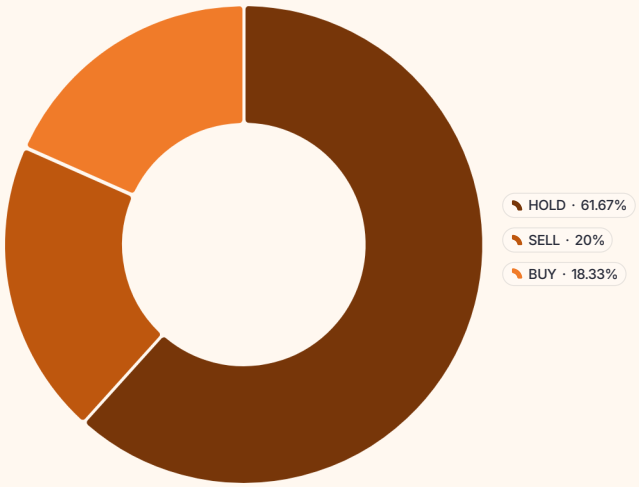
```
if composite_score > +0.31: BUY
elif composite_score < -0.31: SELL
else:                 HOLD

BUY:  SPX=0.40 BCOM=0.15 Corp=0.45 UST=0.00
SELL: SPX=0.05 BCOM=0.00 Corp=0.35 UST=0.60
HOLD: carry previous month's allocation
```

Design Rationale

±0.31 dead-band filters low-conviction trades, selected via grid sweep (0.25–0.50). HOLD carries forward until explicitly overridden.

Mentor Q: 0.31 was grid-optimised. Should it eventually be dynamic as market regimes shift?

Signal Distribution — 360 Months



The model acts less than 40% of the time. Selectivity is a feature — it keeps monthly turnover near 5%, limiting transaction cost drag.

Benchmark - Measure Against 25% Equal-Weight Passive Strategy

We have experimented with 4 different benchmarks, each testing a distinct null hypothesis about its value-add.

Equal-Weight Passive

25% each asset, monthly rebalance. Tests: *does M1 beat naive diversification?*

Buy-and-Hold SPX

100% equity, no rebalance. Tests: *does M1 beat doing nothing in equities?*

60/40 SPX/UST

Classic institutional allocation. Tests: *does M1 beat the industry standard?*

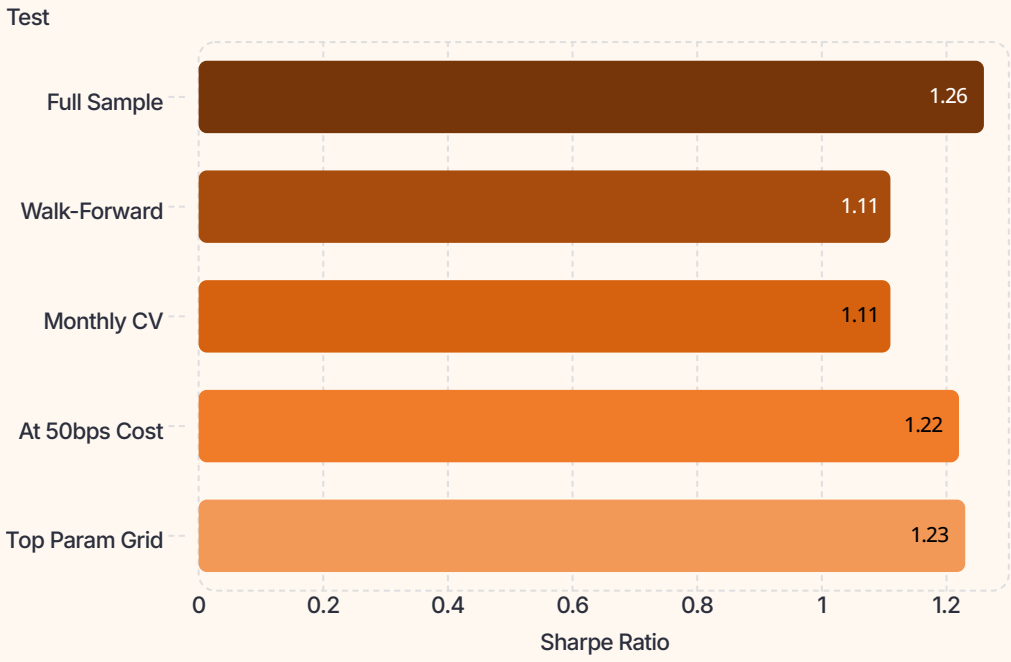
Simple Trend (MA)

10/12-month SPX crossover. Tests: *does M1 add value beyond a single rule?*

Mentor Q: 25% equal-weight is unrealistic institutionally. Would a market-weight or survey-based benchmark be a more honest baseline?

M1 Baseline Performance & Robustness

Full Sample vs Stress Tests



What Held Up

- **Win rate:** 67–69% stable across all test configurations
- **Max drawdown:** −14.4% consistent in walk-forward and CV
- **Transaction costs:** 50bps degrades Sharpe by only 3.2% — low turnover (5%/mo) provides natural protection
- **Threshold sensitivity:** ±0.06 from 0.31 keeps Sharpe above 1.20

What Showed Sensitivity

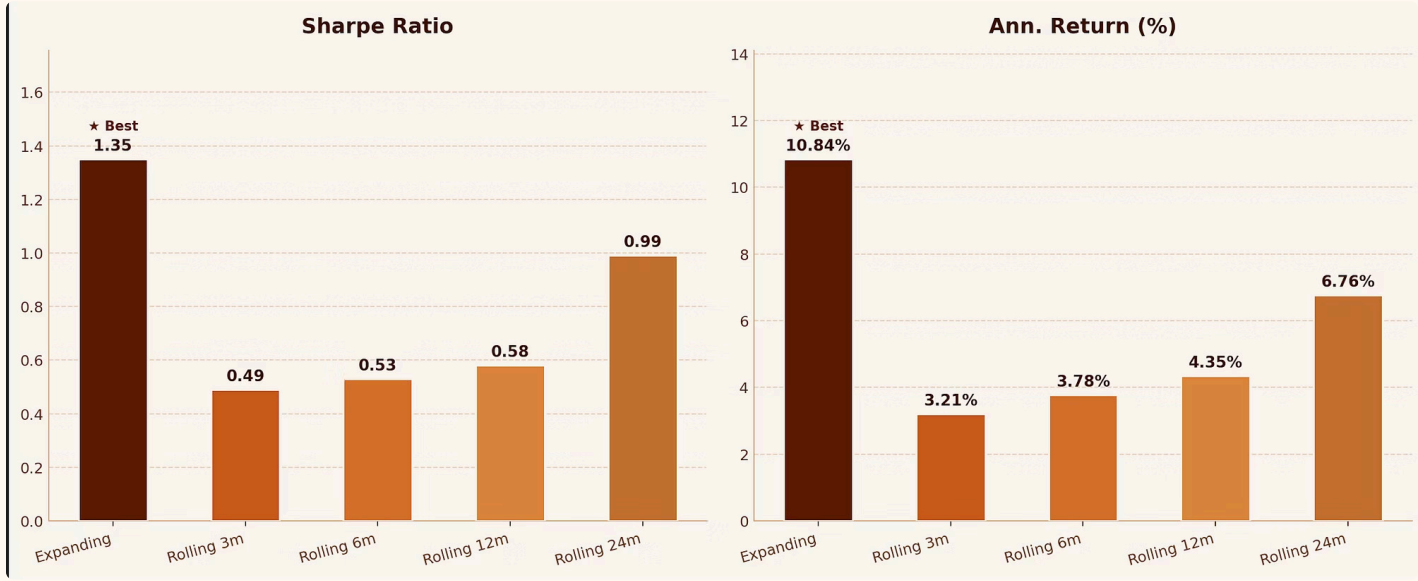
Annualised return drops ~1pp in walk-forward vs full-sample, suggesting the dynamic IC weights capture some in-sample benefit.

Mentor Q: Walk-forward Sharpe drops 12% out-of-sample. Is that normal shrinkage, or a sign the dynamic IC weights are overfitting?

Validation Design: Rolling vs Expanding Windows

Walk-Forward Protocol

```
ROLLING:   train on [t - window, t)
           window ∈ {3, 6, 12, 24, 36}
EXPANDING: train on [start, t)

Results:
  Expanding:    Sharpe 1.11
  36mo rolling: Sharpe ~1.08
  3mo rolling:  Sharpe ~0.85 (unstable)
```



Why Expanding Wins

For macro indicators, the expanding assumption is more defensible — yield curve inversions in 2000 remain informative about inversions in 2023. Short windows discard that structural knowledge, producing noisy z-scores and unstable signals.

Mentor Q: Should the expanding window be soft-capped (e.g., max 15 years) to avoid stale anchoring as early-period data accumulates outsized influence over time?

M2 Framing — What the Meta-Labeling Layer Does

Meta-Label Definition

For each actionable M1 event at time *t*:

- Meta-target = portfolio return from t → t+1
- Meta-label = 1 if return > 0, else 0
- HOLD months excluded — M2 only evaluates moments when M1 acts

Dataset: 138 events (66 BUY + 72 SELL) over 28 years. Base rate: ~70% wins.

Why Binary Classification?

M2's job is accept/reject. Binary classification — 'will M1 be right?' — is tractable at 138 observations; continuous return prediction is not.

The Structural Constraint

138 events over 28 years ≈ 5/year, with only ~41 loss events. This is the hard ceiling on what M2 can learn — hence simple models only.

Mentor Q: 138 events can't be expanded without changing the strategy. Does this fundamentally cap what meta-labeling can achieve here?

Evaluating M2 — Precision, Recall, and the ROC Curve

Why Accuracy Is the Wrong Metric

Approving everything achieves **70% accuracy with zero value-add**. Precision/recall reveal the real tradeoff: higher threshold → fewer false approvals but more good trades missed.

The Financial Asymmetry

Capital-preservation tolerates **low recall**. Return-maximisation cannot miss good trades. ROC-AUC summarises this across all thresholds.

ROC Results

AUC ~0.52 means M2 barely ranks trades better than random. But weak ranking can still help for sizing — even a slight edge avoids binary cash drag.

Mentor Q: AUC 0.52 is indistinguishable from random at n=138. Is this grounds to abandon M2, or a problem with how we're using the probability output?

Feature Engineering — VIX and OAS as Regime Indicators

22 Candidate Features

10 regime features (5 VIX + 5 OAS): level z-score, change z-score, 6mo trend, regime flag, direction flag. 12 M1 context features: composite score, direction, 6 indicator z-scores, trailing hit rate, avg return, vol (all 12mo), signal streak. Trailing features tell M2 how M1 has been performing lately — something M1 itself is blind to.

Hypothesis

Calm markets (low VIX, tight OAS) → stable trends → M1 wins more. Stressed markets → trends break → M1 underperforms.

Mentor Q: Should M2 have been restricted to trailing performance features and regime indicators only, excluding M1's own inputs?

No-Lookahead Enforcement

```
# VIX features (5):
vix_level_z  = expanding_zscore(VIX)
vix_change_z = expanding_zscore(VIX.diff())
vix_trend    = VIX > rolling_mean(VIX, 6mo)
vix_regime   = VIX > expanding_median(VIX)
vix_rising   = (VIX.diff() > 0).astype(int)

# OAS features (5):
oas_level_z  = expanding_zscore(OAS)
oas_change_z = expanding_zscore(OAS.diff())
oas_trend    = OAS > rolling_mean(OAS, 6mo)
oas_regime   = OAS > expanding_median(OAS)
oas_widening = (OAS.diff() > 0).astype(int)

# All z-scores use shift(1):
# current value never included in its
# own normalisation statistics.
```

Every expanding z-score and median uses only observations before time *t*. — no look-ahead contamination.

M2 EDA (Exploratory Data Analysis) — Do VIX and OAS Predict M1 Wins?

All Statistical Tests: Negative

Chi-Square: VIX vs M1 Win

p = 0.8882 — no signal

Chi-Square: OAS vs M1 Win

p = 0.7730 — no signal

T-Test: Returns by VIX Regime

p = 0.1597 — not significant

Max Feature Correlation

+0.108 with meta_label

Mentor Q: M1 peaks during stress — SELL signals sharpen in sharp downturns. Should VIX be an amplifier, not a filter?

The Counterintuitive Result

High-stress environments actually improve M1 performance:

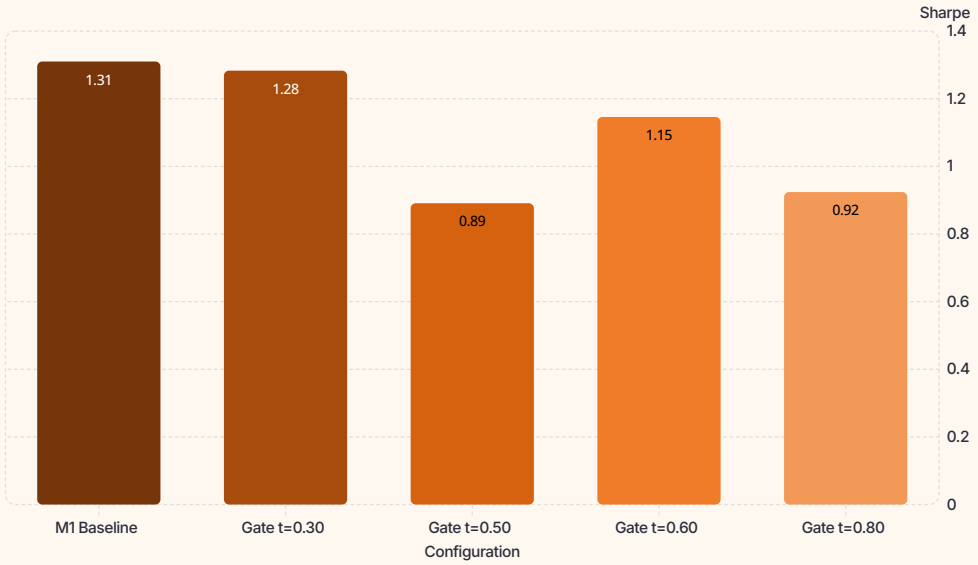
- **High VIX + Wide OAS:** 72.7% win rate, +1.02% avg return — best regime
- **Low VIX + Tight OAS:** 70.6% win rate, +0.31% avg return
- **Regime gate (calm only):** Sharpe collapses from 1.19 → 0.68

Financial Explanation

M1 is trend-following — stress produces larger, more directional moves where trend signals have the clearest edge. Filtering out stress removes M1's best regime. VIX and OAS are not orthogonal to M1's own inputs.

M2 Iteration 1 — Binary Gate (22 Features)

Results — Negative IR at Every Threshold



Why Every Threshold Failed

Rejecting a trade earns zero, but M1 wins 70% of the time. The asymmetry is fatal: you forfeit upside on 70% of rejections, save downside on only 30%. Guarantees negative IR without near-perfect discrimination — impossible at n=138.

Model Choice Rationale

Logistic regression: interpretable, no hyperparameters, appropriate for binary target at small N. Walk-forward with 60-event minimum training window.

Mentor Q: Is the binary gate fundamentally incompatible with a high-win-rate primary model?

M2 Iteration 2 — Feature Selection (Drop VIX/OAS)

Rationale

VIX and OAS carry no independent signal (p=0.89, p=0.77). 10 noise dimensions fragment the decision boundary at n=138. Dropping them reduces variance without losing signal.

Iteration 2 vs. Iteration 1: Marginal Improvement

t=0.30: v2 IR -0.072 vs v1 IR -0.236
t=0.51: v2 IR -0.607 vs v1 (comparable)

AUC: 0.523 → 0.528 (slight improvement)

Cleaner features improved M2's discrimination slightly — but the structural cash-drag mechanism has not changed. The binary gate is still the problem.

Mentor Q: At t=0.30, v2 achieves Sharpe 1.371 (above M1's 1.310) but IR is decreasing by −0.072. Is a slightly worse but higher Sharpe a defensible outcome, or is any negative IR disqualifying?

The Orthogonality Problem

A feature is useful to M2 only if it captures variation in M1's outcome that M1 cannot explain from its own inputs. Most features tried so far fail this test — they are either M1 inputs in disguise or redundant proxies.

Why Current Features Fall Short

VIX

Proxies for risk breadth and credit stress — already captured by M1's own indicators

OAS

Tracks credit risk appetite — directly redundant with M1's credit vs rates signal

M1 Z-Scores

M1 already acted on these — M2 cannot extract new information from M1's own inputs

Mentor Q: Few truly orthogonal features vs. more features with genuine new information — which principle should govern M2 feature selection?

Candidate Orthogonal Features (Next Steps)

Cross-Asset Dispersion (3m)

Return breadth across assets — orthogonal to M1's directional signals.

SPX Realized Vol (3m ann.)

Volatility regime shifts not encoded in M1's z-scores.

Vol-of-Vol (6m std of rvol)

Detects unstable regimes where M1's confidence should be discounted.

Corp Bond Carry (12m)

Income-based credit signal, distinct from M1's price-momentum credit indicator.

Sentiment Divergence

Positioning vs price — flags when consensus contradicts M1's signal.

M1 Calibration Curve

Win rate by composite score decile — M1's historical accuracy by conviction level.

M2 Iteration 3 — Position Sizing

The Mechanism Change

Replace approve/reject with **continuous position sizing**. M2's probability output directly scales exposure. Normalised so average exposure equals M1 — no systematic de-risking, just reallocation of conviction across months.

```
size = m2_prob / mean(m2_prob)
# avg exposure = 1.0× (same as M1)
# Range: 0.35× to 1.72× across events
# No month is zeroed out.
```

Even with AUC ~0.52, if M2's probabilities are **slightly higher for actual wins**, sizing concentrates risk where it pays off — without cash drag.

First Configuration to Beat M1 on All Metrics

+1.3% return, +0.035 Sharpe, +0.46 IR simultaneously. M2's probability output has **ordering value** even with weak absolute AUC.

Mentor Q: Should position sizes be capped (e.g., max 1.5×) to limit concentration risk, or does capping defeat the purpose of letting M2 express conviction?

M1 Baseline

Return: 9.53%

Sharpe: 1.310

IR: 0.64

M1 + M2 Iteration 3

Return: **10.84%**

Sharpe: **1.345**

IR: **0.83**